



GANGA DRAINAGE SYSTEM — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Ganga — Drainage Type & Bangladesh Names

- Ganga river **Antecedent Drainage** ka example hai. Bangladesh mein pravesh karne ke baad Ganga ko 'Padma' naam se jaana jaata hai — yahan Brahmaputra (jise Jamuna kehte hain) isse milti hai, phir Meghna river se milkar Bay of Bengal mein girti hai. Total length **2525 km** hai. [Q1, Q2, Q3, Q4]

2. Sunderban Delta — World's Largest

- Sunderban Delta **Ganga, Brahmaputra aur Meghna** teeno se milkar banta hai — **duniya ka sabse bada delta** hai. Iska **do-tihai hissa Bangladesh mein** hai, baaki India mein. [Q5]

3. Ganga Pollution — Assertion-Reason

- Ganga ek **highly polluted river** hai (Assertion sahi) — lekin iska reason '**pavitrata**' **NAHI** hai (Reason galat). Asli reason: shehri kshetron se aane wala **high organic-load waste** pollution ka root cause hai. [Q7]

4. Gangetic Alluvial Soil Depth

- Gangetic alluvial soil ki depth: **Oldham ke anusaar 4000-6000 meter**, **Glancy ke anusaar ~2000 meter** — exam options mein **6000 meter** sahi maana jaata hai. [Q7]

5. Panch Prayag — All 5 Confluences

- Panch Prayag (5 sacred confluences), source se confluence tak order mein: **Vishnuprayag (Vishnu Ganga + Dhauliganga = Alaknanda banti hai)** → **Nandprayag (Alaknanda + Nandakini)** → **Karnaprayag (Alaknanda + Pindar)** → **Rudraprayag (Alaknanda + Mandakini)** → **Devprayag (Alaknanda + Bhagirathi = Ganga banti hai)**. Isi tarah Tehri mein **Bhagirathi + Bhilangna** ka sangam hota hai. [Q9, Q10, Q11, Q12, Q13, Q15, Q16, Q17, Q18]

⚠ EXAM TRAP — Nandprayag ka sangam Alaknanda-Nandakini se hai — Vishnu Ganga-Mandakini se NAHI (common UPSC trap)

- Devprayag ka literal matlab 'pavitra sangam' hai — **Adi Shankaracharya ne Joshimath (Badrinath ke paas) ko badi peeth sthapit ki thi, Kedarnath ko NAHI** (Kedarnath unka samadhi sthal hai, ye ek common wrong-statement trap hai). [Q8]
- Badrinath Temple **Alaknanda** ke kinare hai, Kedarnath Temple **Mandakini** ke kinare hai — dono alag rivers ke kinare, common confusion. [Q19]

6. Source of Ganga — Gaumukh, Gangotri, Bhagirathi

- Bhagirathi river **Gaumukh (Gangotri Glacier, Uttarkashi district)** se nikalti hai — Shivalinga peak se 18 km door. Ye **Central aur Lesser Himalaya** ko cut karke behti hai, aur Devprayag mein Alaknanda se milkar 'Ganga' ban jaati hai. [Q15, Q22]

7. Alaknanda Tributaries — Dhauliganga, Mandakini

- Eastern Dhauliganga **Kali river** ki tributary hai (Alaknanda ki NAHI) — Kali river Lipulekh Pass ke paas Kalapani se nikalti hai, iski major tributaries Gori, Saryu, Lohavati, Ladhiya hain. [Q14]
- Mandakini river **Chorabari Taal (Kedarnath ke paas)** se nikalti hai, Sonprayag mein Vasukiganga/Son Ganga se milti hai, aur **Rudraprayag** mein Alaknanda se milti hai. [Q13, Q18]

8. Length Comparisons — Ganga vs Other Rivers

- India ki sabse lambi behne wali river **Ganga (2525 km)** hai — **Godavari (1465 km) > Narmada (1312 km) > Mahanadi (851 km)** se zyada, aur **Brahmaputra ya Indus se bhi zyada lambi** (India mein flowing length ke basis par). [Q20, Q21]

9. Left-Bank Tributaries — Gomti, Ghaghara, Gandak, Kosi

- Sirf ek tributary jo **plains se nikalti hai** — **Gomti** (Gomat Taal/Fulhar Lake, Pilibhit, U.P. se). **Prayagraj ke downstream, west-se-east order: Gomti (Ghazipur mein milti hai) → Ghaghara (Saran, Bihar) → Gandak (Sonapur, Bihar) → Kosi (Katihar, Bihar)** — sab left-bank tributaries hain, jo 11 states ko drain karti hain. [Q23, Q24, Q33, Q34]
- Gandak river ko Nepal mein '**Narayani**' ya '**Gandaki**' bhi kehte hain — do streams (Kaligandak + Trishulganga) se milkar bani hai, Dhaulagiri-Everest ke beech Nepal Himalaya se nikalti hai. [Q37]

10. Yamuna — Origin & Right-Bank Tributaries

- Yamuna river **Yamunotri Glacier (Bandar Poonch Peak ke paas)** se nikalti hai — Delhi-Agra ke beech saal ke zyadatar hisse mein '**dead river**' ban jaati hai (Assertion sahi), lekin iska reason **perennial nahi hona**

NAHI hai (Yamuna asal mein perennial river hai) — real cause **domestic aur industrial waste discharge** hai. [Q25, Q27]

11. Chambal-Betwa-Ken System

- Yamuna ki tributaries, **west-se-east order: Chambal → Sindh → Betwa → Ken**. Chambal M.P. ke Dr. Ambedkar Nagar (purana naam Mhow) se nikalti hai, Rajasthan (Sawai Madhopur, Dholpur) se hokar Yamuna mein milti hai. [Q28]
- Son (Ganga ki tributary) **Yamuna mein NAHI girti** — Chambal, Betwa, Ken sab Yamuna ki tributaries hain, Son inme se ek NAHI hai (common trap). [Q26, Q29]
- Betwa river **Vindhyan Hills** se nikalti hai, Hamirpur (U.P.) ke paas Yamuna mein milti hai — **Chambal ki tributary NAHI hai** (ye Yamuna ki alag tributary hai, Chambal mein nahi milti). [Q30, Q31, Q32]

12. Son River — Right-Bank, Northward Flow

- Son river (Amarkantak, M.P. se nikalti hai) **Ganga ki main right-bank tributary hai jo North ki taraf behti hai** — Patna ke paas Ganga mein milti hai; baaki left-bank tributaries South ki taraf behti hain. **Kaimur Range** Son (south) aur Tons/Yamuna-tributary (north) ke beech watershed banati hai. Son ka **catchment area (~70,055 sq km)** **sabse zyada** hai (Gandak, Ramganga, Mahananda se). [Q33, Q34, Q35, Q36]

13. Ganga Sub-Basin Tributaries (Damodar, Koel, Sankh)

- Ganga river basin ke sub-rivers: **North Koel (Son ki tributary), Barakar (Damodar ki main tributary, Koderma Plateau se nikalti hai)** — dono Ganga basin ka hissa hain. **South Koel aur Sankh, Brahmani river system ka hissa hain (Ganga basin ka NAHI)** — Damodar, Hooghly se milkar lower Ganga basin ka sub-basin banata hai. [Q39]
- Ganga basin mein NAHI: **Jonk river Mahanadi ki tributary hai** (Odisha-Chhattisgarh) — **Punpun, Ajoy, Jalangi sab Ganga basin ka hissa hain**. [Q42]
- Gandak river U.P. se NAHI guzarti** — Nepal se seedhe Bihar mein aati hai. **Son, Hindon, Sharda rivers U.P. se guzarti hain**. [Q43]
- Punpun river **Chota Nagpur Plateau (Jharkhand)** se nikalti hai, Aurangabad (Bihar) se hoke **Fatuha (Patna)** mein Ganga se milti hai. [Q44]

14. Ganga in Bihar — Districts & Length

- Ganga Bihar ke **12 districts** se guzarti hai — Saran, Buxar, Bhojpur, Patna, Vaishali, Samastipur, Begusarai, Lakhisarai, Munger, Khagaria, Bhagalpur, Katihar. Inme sabse zyada length **Patna (~99 km)** mein hai — Bhagalpur (~97 km) dusre number par. [Q40, Q41]

15. Moribund Delta & Bengal Delta Stages

- Bengal Delta ke 4 stages hain: **Moribund (south-west, Jessore-Kushtia-Faridpur, highly-silted oxbow lakes) → Immature (Moribund ke south, Sundarbans yahin hai) → Mature (Bangladesh ke southern central section) → Active (Meghna delta, Bay of Bengal mein, regular flooding aur 'chars')**. [Q38]

16. Kosi Plain — Elevation & Course

- Kosi Plain (triangular, Bihar mein, lower Ganga plain ka hissa) — **opening Birpur (54m, Nepal border) par hai, base Khagaria (36m) se Kursela (25m) tak Ganga ke parallel jaata hai, average elevation 30-40m** — general slope northwest se southeast. *[Q45, Q46]*

17. Dendritic Drainage Pattern

- Ganga plains aur Assam Valley mein **Dendritic drainage pattern** milta hai — wide plains, gentle slopes, heavy rain, bahut saari tributaries ki characteristic. Ye pattern un areas mein develop hota hai jahan underlying rocks mein koi specific structure nahi hoti, isliye streams sab directions mein evenly failti hain. *[Q47]*